

9.3 - REPRESENTING RELATIONS

There are many ways to represent a relation between finite sets.

one way : to list its ordered pairs

2nd way : to represent a relation by a table

Two other alternative methods for representing relations .

- **Zero-one matrices** : appropriate for representation of relations in Computer programs
- **Directed Graph** : a pictorial representation, useful for understanding the properties of relations or visual display of information about relation.

REPRESENTING RELATIONS USING MATRICES :

A relation between finite sets can be represented using **zero-one matrix**.
Suppose R is a relation from A to B , where

$$A = \{a_1, a_2, \dots, a_m\} \quad B = \{b_1, b_2, \dots, b_n\}$$

The relation R can be represented by the matrix as :

$$M_R = [m_{ij}] \quad , \text{ where } m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R, \end{cases}$$

i.e., zero-one matrix representing R has

- a 1 as its (i, j) entry where a_i is related to b_j
- a 0 in this position if a_i is not related to b_j

EXAMPLE #1

Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$

Let $R = \{(a, b) \mid a > b, a \in A, b \in B\}$

What is the matrix representing R ?

Sol:

$$A = \{1, 2, 3\} \quad \text{and} \quad B = \{1, 2\}$$

$$R = \{(a, b) \mid a > b\} = \{(2, 1), (3, 1), (3, 2)\}$$

The matrix representation for R :

$$M_R = \begin{bmatrix} (1,1) & (1,2) \\ (2,1) & (2,2) \\ (3,1) & (3,2) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

EXAMPLE # 2

Let $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3, b_4, b_5\}$

which order pairs are in the relation R represented by the matrix,

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Sol:

The relation R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}$$

REFLEXIVE, SYMMETRIC AND TRANSITIVE PROPERTY :

The matrix of a relation on a set, which is square matrix can be used to determine whether the relation has certain properties.

REFLEXIVE :

R is reflexive iff $(a_i, a_i) \in R$ for $i=1, 2, 3, \dots, n$

i.e.,

R is reflexive if all the elements on the main diagonal of M_R are equal to 1.

$$M_R = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$

SYMMETRIC :

The relation R on the set $A = \{a_1, a_2, \dots, a_n\}$ is symmetric iff

$$(a_j, a_i) \in R \text{ whenever } (a_i, a_j) \in R$$

In terms of the entries of M_R ,

R is symmetric iff $m_{ji} = 1$ whenever $m_{ij} = 1$

this also means $m_{ji} = 0$ whenever $m_{ij} = 0$

i.e.,

R is symmetric iff $m_{ji} = m_{ij}$ for all pairs of integers i and j

$$M_R = (M_R)^t \quad : \text{Definition of transpose}$$

i.e.,

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

ANTI-SYMMETRIC :

The matrix of an antisymmetric relation has the property that

$$\text{if } m_{ij} = 1 \text{ with } i \neq j \text{ then } m_{ji} = 0$$

i.e.,

$$\text{Either } m_{ij} = 0 \text{ or } m_{ji} = 0 \text{ when } i \neq j$$

The form of matrix for an antisymmetric relation is shown as:

EXAMPLE #3

Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Is reflexive, symmetric and/or antisymmetric?

Sol:-

Reflexive:

Because all the diagonal elements are equal to 1, so R is reflexive.

Symmetric:

As here $m_{ij} = m_{ji}$, R is symmetric as M_R is symmetric

Antisymmetric:

For antisymmetric, we have to show

$$\text{if } m_{ij} = 1, \text{ then } m_{ji} = 0 \text{ (} i \neq j \text{)}$$

or

$$\text{Either } m_{ij} = 0 \text{ or } m_{ji} = 0 \text{ (} i \neq j \text{)}$$

R is antisymmetric as M_R is antisymmetric.

COMBINING RELATIONS USING MATRICES:

EXAMPLE #4

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices M_{R_1} and M_{R_2} :

$$M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Sol:

UNION: Either M_{R_1} or M_{R_2} has a 1

$$M_{R_1} \cup M_{R_2} = M_{R_1} \vee M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

INTERSECTION: Both M_{R_1} and M_{R_2} have a 1

$$M_{R_1} \cap M_{R_2} = M_{R_1} \wedge M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

COMPOSITE OF RELATIONS:

The matrix can be found using the Boolean product of the matrices for these relations. Suppose R is a relation from A to B and S is a relation from B to C .

Suppose that A, B, C have m, n, p elements.

Let the zero-one matrices for

$$R \text{ be } M_R = [r_{ij}]_{m \times n}$$

$$M_S = [s_{ij}]_{n \times p}$$

$$M_{S \circ R} = [t_{ij}]_{m \times p}$$

The ordered pair $(a_i, c_j) \in S \circ R$ iff there is an element b_k such that $(a_i, b_k) \in R$ and $(b_k, c_j) \in S$.

It follows that

$$t_{ij} = 1 \text{ iff } r_{ik} = s_{kj} = 1 \text{ for some } k$$

From the definition of the Boolean product

$$M_{S \circ R} = M_R \odot M_S$$

EXAMPLE # 5 :

Find the matrix representing the relation $S \circ R$, where the matrices representing R and S are :

$$M_R = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad \text{and} \quad M_S = \begin{matrix} & \begin{matrix} c_1 & c_2 & c_3 \end{matrix} \\ \begin{matrix} b_1 \\ b_2 \\ b_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$R = \{(1,1), (1,3), (2,1), (2,2)\}$$

$$S = \{(1,2), (2,3), (3,1), (3,3)\}$$

Sol :

$$M_{S \circ R} = M_R \odot M_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

R	S
1 → 1	1 → 2
1 → 3	2 → 3
2 → 1	3 → 1
2 → 2	3 → 3
S ∘ R	
1 → 2	
1 → 1	
1 → 3	
2 → 2	
2 → 3	

The matrix representing the composite of two relations can be used to find the matrix M_R^n .

In particular

$$M_R^n = M_R^{(n)}$$

EXAMPLE # 6

Find the matrix representing the relation R^2 , where the matrix representing R is

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Sol :

The matrix R for $_2$ is $R^2 = R \circ R$

$$M_R^2 = M_R^{(2)}$$

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$M_R^2 = M_R^{(2)} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

R^2 is the composite of R with itself and it consists of ordered pairs (a,c) such that $(a,b) \in R, (b,c) \in R$.

$$R = \{(1,2), (2,2), (2,3), (3,1)\}$$

R	R	R ∘ R
(1,2)	(1,2)	(1,2) -
(2,2)	(2,2)	(1,3) -
(2,3)	(2,3)	(2,2) -
(3,1)	(3,1)	(2,3) -
		(3,1) -
		(3,2)

REPRESENTING RELATIONS USING DIGRAPH :

Each element of the set is represented by a point and each ordered pair is represented using an arc with its direction indicated by an arrow.

DEFINITION :

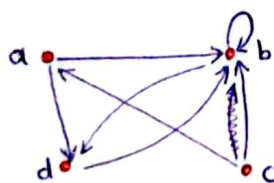
A directed graph or digraph consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs). The vertex a is called the initial vertex of the edge (a, b) , and the vertex b is called the terminal vertex of this edge.

NOTE :

An edge of the form (a, a) is represented using an arc from vertex a back to itself is called a loop.

EXAMPLE #1

The directed graph with vertices a, b, c and d and edges $(a, b), (a, d), (b, b), (b, d), (c, a), (c, b), (d, b)$, is displayed as



The relation R on a set $A = \{a, b, c, d\}$ is represented by directed graph that has the elements of A as its vertices and ordered pairs (a, b) , where $(a, b) \in R$ as edges. This sets up one-to-one correspondence between the relations on a set and all directed graphs with A as their set of vertices.

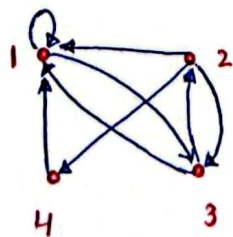
EXAMPLE #2

Build the directed graph of the relation

$$R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$$

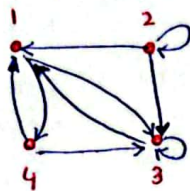
on the set $\{1, 2, 3, 4\}$

Sol :



EXAMPLE #3

What are the ordered pairs in the relation represented by the directed graph shown below:



Sol: The ordered pairs (x, y) in the relation are:

$$R = \{(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 3)\}$$

NOTE:

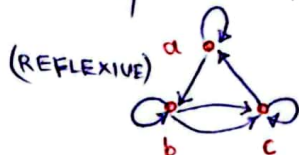
We use such pictorial representation when we think of the relations on a finite set as directed graphs or digraphs.

The directed graph representing a relation can be used to determine whether the relation has various properties.

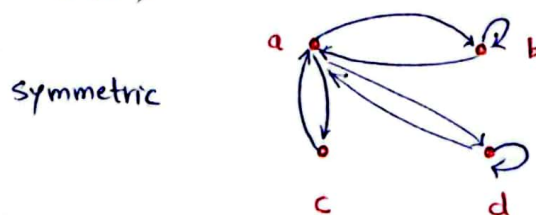
For instance,

REFLEXIVE:

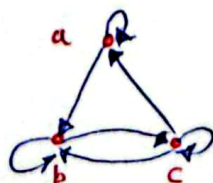
A relation is reflexive **iff** there is a loop at each vertex of the directed graph so that every ordered pair of the form (x, x) occurs in relation.

**SYMMETRIC:**

A relation is symmetric **iff** for every edge between distinct vertices in its digraph, there is an edge in the opposite direction, so that (y, x) is in the relation whenever (x, y) is in the relation.

**ANTISYMMETRIC:**

A relation is antisymmetric **iff** there are never two edges in opposite direction between distinct vertices.

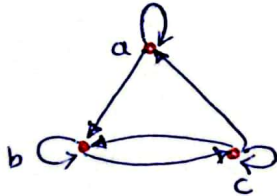


neither symmetric nor antisymmetric.
(nor transitive)

Transitive :

A relation is transitive **iff** whenever there is an edge from a vertex x to a vertex y and

an edge from vertex y to vertex z , there is an edge from x to z (completing a triangle where each side is a directed edge with correct direction).



not transitive as

there is an edge from a to b and b to c but no edge from a to c .

NOTE :

A symmetric relation can be represented by undirected graph, which is a graph where edge donot have directions.

NOTE :

The relation from set A to a set B can be represented by a directed graph where there is a vertex for each element A and a vertex for each element of B will be show in next chapter.

However, when $A=B$ (A to itself), such representation provides much less insight than the digraph representation described here.